TOMMASO CRIPPA

High Performance Computing Engineering Graduate

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EDUCATION

Double Master's Degree in High Performance Computing — **Politecnico di Milano - University of Luxembourg**

110 cum laude / 110 — 2024 - 2026

- Selected for the exclusive EUMaster4HPC double-degree programme, in which I participated in extracurricular HPC challenges, an international summer school and workshops with industry leaders

Bachelor degree in Computer Engineering — **Politecnico di Milano**

110 cum laude / 110 — 2021 - 2024

- Won "Migliori Matricole" scholarship for being one of Politecnico di Milano's best freshmen students of the 21/22 season.

Relevant courses completed: Parallel Computing, Advanced Computer Architectures, High Performance Data Analytics and Visualisation, Programming Machine Learning Algorithms for HPC, Artificial Neural Networks and Deep Learning, Time Series Analysis

EXPERIENCES

Master's Thesis Intern @ MEGWARE Computer Vertrieb und Service GmbH — **Chemnitz**, Germany

HPC | Machine Learning | Time-Series | SLURM | Linux — February 2026 - August 2026

Crippius/hpc-bottleneck-detector

- Developed a diagnosis pipeline analyzing 100+ HPC hardware-counter metrics at ∼5,000 data points per minute across compute-bound, memory-bound, and load-imbalance categories, without source-code instrumentation.
- Validated against HPAS fault-injection ground truth and generalized well to an unseen architecture with no retraining, reaching 0.954 macro-F1 in leave-one-out cross-validation.

EUMaster4HPC Summer School on HPC for Specialised Applications — **Ankara**, Turkey

HPC | CFD | Parallel Programming | NLP — July 2025

- Attended the EUMaster4HPC Summer School at METU, covering parallel programming, CFD simulations, and AI across aeronautics, materials science, and oceanography, with hands-on industry training from MathWorks and other HPC specialists.

HPC Ambassador @ EUROHPC Summit — **Krakow**, Poland

HPC | AI | Fluid Dynamics — March 2025

- Represented the EUMaster4HPC programme at the EuroHPC Summit, delivering live demonstrations of fluid dynamics simulations and LLM deployment on two of the biggest EuroHPC supercomputers (Leonardo and Discoverer) to the general public and HPC community.

Operations & Logistic @ Google Developer Group On Campus - Polimi — **Milan**, Italy

AI | Cloud | Mobile — August 2024 - August 2025

- Organised and executed 5+ technical events over 12 months on AI, Cloud Computing, and Mobile Development, including a hackathon with 150+ participants, doubling community membership over the year.

PROJECTS

Early Alzheimer Detection with Deep Learning

Python | pytorch | keras | nilearn | C++ | libtorch | clinica | SLURM — justvicthor/Early_Alzheimer_Detection_DL

- Trained a 3D CNN for Alzheimer's classification on the ADNI dataset using the MeluXina EuroHPC supercomputer, implementing SLURM-scheduled preprocessing pipelines and GPU-parallelised training achieving Macro-AUC of 0.9197, up from 0.80 baseline.

High-Performance Graph Coloring with Branch-and-Bound Framework

Python | MPI | SLURM — Rudolfovoorg/EuroHPC_Student_Challenge_2025_Team_3

- Won the EuroHPC Student Challenge 2025 by parallelising a Branch-and-Bound solver with a load-balanced MPI manager/worker architecture, scaling to 1024 cores on VEGA to solve over half of the provided benchmark graphs optimally.

GraphSAGE for Large-Scale Node Classification

Python | pytorch | PyG | SLURM | CUDA | DDP — albertofinardi/Sage_GNN

- Trained GraphSAGE on ogbn-products (2.4M nodes, 61M edges) on MeluXina with PyTorch DDP, achieving 3.73× intra-node speedup (93% efficiency) across 4 A100 GPUs, and pruning final-layer neighbor sampling to cut epoch time 2.7× beyond an already paper-tuned baseline, across 36 tested configurations.

HPC Streaming Telemetry Pipeline

</> **Python** **Kafka** **Spark** TimescaleDB Docker Grafana Crippius/hpc-streaming-telemetry

- Built a Kafka and Spark Structured Streaming pipeline over Marconi100 supercomputer telemetry (6.3M events), sustaining 74k messages/s at 3.3 s median end-to-end latency, and evaluated online node-drain detection and 1-hour anticipation on 146 nodes over 2.5 years, with exactly-once windows across a Spark restart.

SKILLS AND LANGUAGES

Tech Stack: Python (Advanced), C, C++, Java, Spark
Infrastructure: Git (Advanced), SLURM (Advanced), DVC, CUDA
Languages: Italian (Native), English (Proficient – IELTS 7.5), German (Beginner – A2)